W-B-137C
April 4, 1973
SUPERSEDING
Fed. Spec. W-B-137A
April 18, 1966, and
Int. Fed. Spec. W-B-00137B(GSA-FSS)
September 30, 1970

FEDERAL SPECIFICATION

BATTERY, STORAGE, (NICKEL ALKALINE, INDUSTRIAL, FLOATING SERVICE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers industrial storage batteries of the nickel-alkaline type for stationary service with float or trickle charging.

1.2 Classification.

1.2.1 Types. Individual storage batteries, as covered by this specification, shall be of the voltage, ampere-hour capacity, and within the dimensional limitations, as specified in the invitation for bids (see 6.1). They shall be of the following types, as specified (see 6.1):

Type I - Nickel cadmium.
Type II - Nickel iron.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification.

Federal Specifications:

O-B-41 - Battery Water.
PPP-B-140 - Batteries, Storage, Industrial: Preparation for Shipment and Storage of.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. All materials used in the construction of batteries purchased under this specification shall equal or be superior to the manufacturer's best commercial established standards. Unless otherwise specified (see 6.1), the positive active material shall be essentially nickelous or nickelic hydroxide and the negative active material shall be iron or cadmium oxides. Additives to improve the performance of the active materials are permissable. The electrolyte shall be an aqueous solution of potassium hydroxide. The battery shall be new, that is, not previously used. Specifically, it shall contain no plates, rubber, plastic, glass, nor other components previously used in other batteries, and shall be free from reclaimed active material.

3.2 Plates.

3.2.1 Positive plates. The positive plates shall be of the multitubular type or perforated pocket type.

3.2.2 Negative plates. The negative plates shall be of the perforated pocket type and shall be of the size and number that conform to the manufacturer's commercial standards for the various sizes and types of batteries. The negative plates shall approximate the sizes of the positive plates in length and width.

3.3 Plate frames. The plate frames of all batteries shall be identical to the manufacturer's commercial product of alkaline types.

3.4 Plate groups. Plates of like polarity in each cell shall be assembled to their respective pole groups either by bolted connections through eyes at the top of the plates or by welding the tops of the plates into a comb on the base of the pole. Bolted or welded connections to the pole shall provide adequate electrical conductivity and mechanical support for the plate groups.

3.5 Elements and separators. Elements shall be assembled by intermeshing the positive and negative groups. The groups shall be insulated from each other by rubber or plastic rods, pins, or sheets, or by ribs molded in the jar container.

3.6 Electrolyte. Electrolyte furnished in or with the cells shall be a solution of potassium hydroxide in water. The water shall be in accordance with Federal Specification O-B-41c. A small amount of lithium hydroxide is permissible as an addition when required. Impurities in the form of carbonates shall be limited to 5 percent, measured in percentage of total alkalinity. When so specified, dry electrolyte (powder) shall be furnished in lieu of liquid (see 6.1). The electrolyte supplied in or with the above cells, shall have a specific gravity of 1.180 to 1.220 for nickel-iron-alkaline batteries and 1.160 to 1.230 for nickel-cadmium-alkaline batteries.

3.7 Posts and intercell connector. Posts and intercell connectors shall be adequate to carry, without excessive heating or voltage drop, the current incident to the service applications. The posts shall be nickel-plated steel, tapered or threaded, internally or externally, with a standard thread to accommodate a locknut. The intercell connectors shall be nickel-plated steel or copper of either the flat strap type or the rod and tapered lug type depending upon the type of pole used.

3.8 Terminal markings (polarity). Polarity of the positive cell terminal shall be plainly and permanently marked "POS", "P", "+", or the post bushing shall be red. The negative terminal shall be marked "NEG", "N", "-", or the post bushing shall be black or blue.

3.9 Cell containers and covers. Cell containers and covers shall be not less than No. 22 gauge nickel-plated steel or of an alkali-resistant plastic, as specified (see 6.1). Plastic material shall be translucent and show no cracks or blisters and shall neither increase more than 1.5 percent in weight, nor 2.0 percent in any dimension when tested as specified in 4.5.4. Steel cell containers and covers shall be joined together by welding to form a sealed unit. Plastic cell containers and covers shall be joined together by cementing to form a sealed unit. Units shall be so designed as to permit quick inspection and the adding of water or electrolyte when required.

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3.10 Assembly. Cells shall be furnished as individual cells or complete batteries, as specified by the purchaser (see 6.1).

3.11 Charge condition for shipment. As specified by the purchaser (see 6.1), cells or batteries, shall be shipped in one of the following conditions:

(a) Charged-and-wet.

(b) Discharged and short circuited for Nickel-Iron types for indefinite storage and for Nickel-Cadmium types for storage up to 6 months.

(c) Short circuited and dumped for overseas shipments of both Nickel-Iron and Nickel-Cadmium types and for Nickel-Cadmium types to be stored longer than 6 months.

If not specified, batteries shall be shipped charged-and-wet ready for service.

3.11.1 Charged-and-wet. Charged-and-wet is the condition of a storage battery when it is filled with electrolyte and is fully charged before shipment from the factory.

3.11.2 Discharged-and-short circuited. Discharged-and-short circuited is the condition of a storage battery which is filled with electrolyte but has been discharged and short circuited. Removal of short-circuiting wires or straps and an initial charge are required before it is ready for use.

3.11.3 Short-circuited-and-dumped. Short-circuited-and-dumped is the condition of a battery which has formed plates which have been discharged and short-circuited before draining of the electrolyte. Filling with electrolyte and an initial charge are required before it is ready for use.

3.12 Weights. Average weights of cells and battery assemblies shall approximate the manufacturer's published catalog values. No extraneous material or parts shall be incorporated to add excess weight to cells or battery assemblies.

3.13 Markings. Batteries or cells furnished as individual units shall be legibly and permanently marked with at least the following:

(a) Name or registered trademark of the manufacturer.

(b) Manufacturer's type designation including cell type.

(c) Ampere-hour capacity and hour rating.

(d) Month and year of manufacture in arabic numerals (eg., 5-70).

Markings may be embossed or indented on the cell cover, cell container (except the bottom), top of terminal posts or connector, or a marking plate may be permanently attached to the battery or rack.

3.14 Instructions. With each shipment of batteries, there shall be included, for each type and size of cell in the shipment, a set of complete instructions for the proper installations and maintenance of the battery including charge and discharge characteristics; it shall state pertinent data regarding identity of the battery with relation to type, size, and manufacture. The instructions shall be securely fastened in plain view of the top of the battery, properly protected from mechanical or chemical damage.

3.15 Performance.

3.15.1 Capacity ratings. When conditioned and tested as set forth in 4.4 and 4.5, individual cells or storage batteries made from individual cells shall give initially not less than 90 percent of the ampere-hour capacity specified (on a time basis) and shall be of such design that when operated on a full-float routine, compatible with economical use of the battery under service conditions, it will consistently attain its rated capacity within five years.

3.15.2 Life. The battery shall be of a design for which test data can be made available by the manufacturer to show that batteries of like design have been on test (see 4.5.2), for at least ten years and that the condition of the test battery at the present time would indicate that the battery will reach a useful life of at least 20 years. The end of service life shall be the time when the battery will no longer deliver 80 percent of its capacity rating as required by 3.15.1.

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3.15.3 Retention of charge. Batteries of nickel-iron construction shall not show a loss of more than 40 percent of their voltage charge while those of nickel-cadmium construction shall not show a loss of more than 15 percent of their voltage charge when tested as set forth in 4.5.3.

3.16 Workmanship. The workmanship in fabricating the parts, the assembly of the cells and battery shall be first class in every particular. The batteries shall be processed in such a manner as to be uniform in quality and free from defects that will affect their life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Where it is impracticable to complete any test on batteries taken from current production prior to delivery to the Government, such as "Life test: (4.5.2), the supplier shall make available his records of tests on preceding production, as necessary to establish proof that batteries of like design, manufactured under substantially identical or essentially equivalent conditions, have met the requirements of this specification.

4.2 Lot. Submission of the product and the formation of lots shall be in accordance with MIL-STD-105.

4.2.1 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105.

4.2.2 Visual, dimensional, and primary functional examination. Each battery or cell, selected in accordance with 4.2.1, shall be inspected for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.2.3 Classification of defects. For examination purposes, defects shall be classified as follows:

Major defects (1.0 AQL)

Material (see 3.1).

Plates (see 3.2).

Positive and negative groups (see 3.4).

Assembly of elements and separators (see 3.5).

Electrolyte (see 3.6).

Posts and intercell connectors (see 3.7).

Polarity (see 3.8).

Cell containers and covers (see 3.9).

Assembly of cells (see 3.10).

Weight of cells (see 3.12).

Major defects (1.0 AQL)

Performance requirements (see 3.15).

Workmanship (see 3.16).

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Minor defects (2.5 AQL)

Charge condition (see 3.11).

Marking (see 3.13).

Missing instructions (see 3.14).

4.3 Sampling for test. Sampling for testing shall be in accordance with table I.

TABLE I. Testing

Group	Tests	Sampling	Acceptance number
I	4.5.4	Level II, MIL-STD-105.	AQL 1.5
II	4.5.1 and 4.5.3	One battery or cell as applicable per month or one for each 500 produced whichever occurs first, except that not more than 3 batteries need be tested in any single month.	0
III	4.5.2	Not less than one battery per year.	0

4.4 Conditioning of samples. If the battery is supplied in a charged-and-wet condition, the conditioning charge prior to the capacity test shall consist of charging for three hours at the 5-hour rate. If the battery is supplied in a short-circuited-and-dumped condition or discharged-and-short-circuited condition, it shall be charged in accordance with the manufacturer's instructions. The charging rate shall be not less than 90 percent of the 5-hour discharge rate and shall be continued until the voltage and current remain constant for three successive hourly readings. The electrolyte temperature during the charging period shall be not less than 60°F. nor more than 115°F. The electrolyte level at the start of charge shall be in accordance with the manufacturer's instructions.

4.4.1 Electrolyte condition. The specific gravity of the electrolyte measured before watering prior to charging, when corrected for proper level and temperature in accordance with the manufacturer's instructions, shall be not less than 1.180 and not more than 1.220 for nickel-iron-alkaline batteries and not less than 1.160 nor more than 1.230 for nickel-cadmium-alkaline batteries.

4.5 Test procedures.

4.5.1 Capacity test. The capacity test shall be started not less than 12 hours and not more than 24 hours following termination of the conditioning charge. The ambient temperature during the test shall be not less than 72°F. nor more than 82°F. The electrolyte temperature shall be not less than 72°F. nor more than 82°F. at the start of discharge. To determine conformance to 3.15.1, batteries shall be discharged at the 8-hour rate to an end voltage of 1.10 volts per cell.

4.5.1.1 If the initial electrolyte temperature is other than 77°F., the discharge time shall be corrected to the standard test temperature of 77°F. by applying a correction factor of 2.2 percent for nickel-iron-alkaline batteries and 2.0 percent for nickel-cadmium-alkaline batteries for every 10°F. of temperature difference. If the initial discharge temperature is above 77°F., the observed time shall be decreased by the above amount; if below 77°F., it shall be increased.

4.5.2 Life test. Cells or batteries shall be placed on float charge at the terminal voltage specified by the manufacturer and inspection records maintained throughout the test period to substantiate conformance with 3.15.2. Range of voltage fluctuations shall be as specified by the manufacturer.

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4.5.3 Retention of charge test. The cell or battery shall be capable of meeting the requirement of 3.15.3 without further charging after being stored in the charge condition for 12 days at an ambient temperature of 80°F. ± 5°F. and a relative humidity of 50 ± 5 percent.

4.5.4 Alkali absorption test. When cell containers and covers are made of material other than steel, two specimens of material, each 3 by 3 inches, shall be cut from the cell container or from cell container material. The cut edges shall be neither polished nor sealed. After being measured with vernier calipers and weighed in the dry condition at 75°F. on a chemical balance, each specimen shall be immersed in a covered vessel containing 150 cc of potassium hydroxide of 1.300 ± 0.050 specific gravity (at 72°F.). This vessel shall be maintained for 7 days in an oven at 150° ± 4°F. At the end of the heating period, the specimens shall be rinsed in water and dried on the surface. They shall then be inspected for evidence of cracks or blisters and shall be measured and weighed. The percentage change in dimensions and weight shall be calculated to determine conformation to the requirements specified in 3.9.

4.5.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-140, except Civil Agency marking shall be in accordance with 5.2.1. The preservation and packaging shall be level A or C and packing shall be A, B, or C (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type of battery required (1.2).

(c) Ampere-hour capacity and voltage (1.2).

(d) Maximum dimensions of battery (see 1.2), and where applicable, a sketch showing space available and terminal arrangement.

(e) Plate materials, if other than specified (3.1).

(f) Specific gravity of electrolyte, if other than specified (3.6.1).

(g) Cell container material (3.9).

(h) Cell or cell assembly desired (3.10).

(i) Charge condition of battery (3.11).

(j) Inspection responsibility, if other than specified (4.1).

(k) Special marking when required (5.2).

(l) Levels of preservation and packaging and level of packing required (5.1).

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Civil agency coordination activity:

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